

CLEANER CHEMICALS

Tata Chemicals is transitioning towards cleaner energy sources, with a focus that has evolved from conserving energy to natural resources

BY KRISHNA GOPALAN

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TATA CHEMICALS IS one of the largest producers of soda ash globally, with a manufacturing footprint across four continents. For the uninitiated, soda ash is chemically known as sodium carbonate and is an inorganic compound used in a host of applications, including manufacturing soaps and detergents, glass, and in metallurgy. Apart from this, the company makes salt and specialty products, which include crop protection and seeds. It has been recognised as Most Sustainable Company in Chemicals in the category of Sectoral Excellence in Manufacturing in this year's *BT* India's Most Sustainable Companies.

In his compact room, Narasimha Kamath V, Tata Chemicals' Chief Manufacturing Officer, speaks of how sustainability "has been integral to the strategy since its inception." He has spent over three decades with the organisation, with almost all that time at the Mithapur plant (in Gujarat), before moving to Mumbai.

Tata Chemicals' history goes back to 1939, and Kamath picks out three phases in the sustainability journey. The first one was when

the entire Okhamandal region faced a drought, and the company adapted its operations by replacing fresh water with seawater. "During the middle years, we built this approach to design our own multi-effect evaporators to process brine (highly concentrated salt mixed with water), to produce water and salt," he explains. This facility came to be known as the make-up water plant, where water is treated as the primary product and salt is the by-product. While water was used for steam generation, the salt was iodised and, over time, became a key product for Tata Chemicals. "Since 2015, we have not used groundwater or surface water from the lake for our process needs," adds Kamath.

During the 1990s, Tata Chemicals developed a patented technology to filter solids from wastewater generated by the soda ash plant. These were then used as raw material for the cement operation. "This plant was specifically designed to utilise rejected limestone from soda ash production, fly ash from the cement plant and filter solids from soda ash wastewater, creating a highly circular system. This level of circularity is rare and distin-



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Chief Manufacturing Officer, Tata Chemicals

MOST SUSTAINABLE COMPANY IN MANUFACTURING (CHEMICALS)

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Moving to cleaner sources requires adapting existing boilers for biomass, converting steam-driven equipment to run on renewable electricity

